EGRE 245 Engineering Programming
Fall 2025
Mark Schwitzerlett

Lab Exercise #11: Bits To String

1 Overview

Write the C function `bitsToStr` that is passed an unsigned integer and returns the value converted to binary represented as a string.

- You must use bitwise operations to extract every bit which will then be stored in the appropriate location in a char array and then returned.
- Make sure the end of the string is marked with a null character!
- Note that you will need to `malloc` the memory for the string that you return. Also you should write your code to be general enough that it would handle unsigned ints of any size.
- Here is code to get you started.

```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>
#define NUMBITS (sizeof(unsigned)*8)
char *bitsToStr(unsigned x) {
char *s = malloc(NUMBITS+1);
// your code goes here
return s;
}
```

2 Code Documentation

The very first lines in *all* of your programming project files this semester should be comments in the class header file (copy from header.c). Of course, the items in the square brackets [] in the comments should be filled in with the proper information.

In addition, the first output for all of your projects this semester, both for the lab exercises and the programming assignments, should be the name of the project and your name, e.g:

```
printf("Lab #11‘ - Joe Student\n");
```

Obviously, for the lab exercises, the output should include the names of both students in the lab group, e.g:

```
printf("Lab #11 - Joe Student and Jane Doe\n");
```

3 Sample Run

Your output should match the sample run below.

```
Mark Schwitzerlett and Jacobi
Lab 12 -- November 2025

size of unsigned int = 4 bytes
x? 263
x = 00000000000000000000000100000111


...Program finished with exit code 0
Press ENTER to exit console.
```

4 Deliverables

You should turn in a stand-alone, complete application program (your C source code, a single ".c" file) containing a `main()` function. All labs and projects this semester will be submitted in Gradescope (via a Canvas assignment).

Your file should be named Lab#11_FirstName_LastName.c

Due date: The lab is due at the end of your lab period, either Tuesday, November 18th or Thursday, November 20th.

Be sure to save a copy of your .c file.